databricks

Big Book of AI Governance

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01

Introduction to AI Governance on Databricks

AI is transforming every part of the enterprise. Modern organizations are embedding AI across customer experiences, software development, business operations and decision-making. As AI evolves from chatbots to fleets of intelligent agents that can reason, retrieve enterprise knowledge, use tools and take action, it creates an unprecedented opportunity to improve productivity, accelerate innovation and create entirely new ways of working.

But AI also changes the rules of governance. Traditional governance has been focused on protecting and managing data. AI introduces a fundamentally different challenge: governing intelligent systems that continuously interact with enterprise data, models, tools and applications. Data teams must ensure AI has the right business context, while developers increasingly rely on coding agents that can access code, tools and enterprise systems, and take actions on their behalf.

Organizations need to ensure these interactions operate within defined policies, protect sensitive information, control rapidly rising costs and remain flexible as new models, agents and technologies emerge. Rather than a standalone checkpoint, governance must be a continuous capability that spans every AI interaction.

This book explores what modern AI governance looks like in practice. You'll learn why governing data and AI together is essential for building trusted AI, the challenges organizations face as they move AI into production and the architectural principles that enable AI to scale securely and responsibly. Through the themes of cost, choice and control, you'll see how leading organizations create a unified foundation for enterprise AI — one that empowers innovation while maintaining trust, security and operational excellence.

What is AI governance?

AI governance is the framework of policies, processes and technical controls that helps you build, deploy and operate AI responsibly. The goal is straightforward: ensure AI systems are trustworthy, secure, compliant and cost-effective while enabling innovation at enterprise scale.

The scope of AI governance has expanded dramatically. A few years ago, governance meant managing model training data, tracking model versions and ensuring fairness. Today, you need to govern much more:

- **Data and business context that AI systems reason over**
- **Models (both proprietary and open source) running across multiple providers**
- **Agents that autonomously retrieve information, invoke tools and take action**
- **MCP servers and tools that connect agents to enterprise systems**
- **Coding agents that write, test and deploy code on behalf of developers**

Most critically, you need to govern the interactions between all of these at runtime, because that is where the real risk and the real value live.

Your organization needs to enable AI adoption while maintaining appropriate security, oversight, accountability and flexibility. Effective AI governance rests on five core principles:

1. **Trustworthy data and context**
2. **Security and policy enforcement**
3. **Transparency and accountability**
4. **Operational oversight**
5. **Cost optimization**

Get these principles right and you can scale AI with confidence. Get them wrong and you face runaway costs, security gaps, compliance failures and eroding trust.

Why organizations need an AI governance framework

Traditional governance was built for a world where data sat in storage and humans decided what to do with it. Access controls, privacy policies, quality checks, lineage tracking and compliance audits worked well when the primary risk was unauthorized access to data at rest.

But in 2026, AI changes the equation. Models and agents don't just read data. Instead, they reason over it, invoke tools, interact with external systems and take actions on behalf of users. Every one of these interactions introduces governance risk that traditional approaches were never designed to handle.

As a result, you're simultaneously dealing with several compounding challenges:

Rapidly growing and difficult-to-attribute AI spend — Every model call, reasoning step and tool invocation has a cost. As agents become more autonomous, token consumption grows in ways that are hard to predict and harder to attribute.

An expanding landscape of proprietary and open models — New frontier models emerge constantly. Your teams want access to the best model for each task, but every new provider means another contract, credential set and governance gap.

More autonomous agents and coding tools — Gartner predicts that by 2028, the average Fortune 500 enterprise will manage more than 150,000 AI agents. Each one can query systems, access sensitive data, call external APIs and make decisions autonomously.

New security, privacy and observability requirements — When an agent accesses customer data, invokes a tool that modifies a production system or generates code that gets deployed, you need to know what happened, why and whether it complied with your policies.

The central governance challenge is clear: You need to govern the data and context AI uses, the AI assets themselves and what AI does at runtime. Traditional governance covers the first. Modern AI governance must cover all three.

Unify data and AI governance on the Databricks Platform

Effective AI governance can't operate in a silo.

AI depends on enterprise data and business context. Your agents need to know which table is authoritative, what "revenue" means in your organization and which documents are trustworthy. If you govern data in one system and AI in another, you create inconsistent policies, duplicated effort and gaps that widen as adoption scales.

Databricks takes a different approach. Instead of bolting everything onto a separate stack, data governance is natively built into your data platform and extended for enterprise data to cover models, agents, tools and runtime AI interactions.

The architecture has two components:

- **Unity Catalog:** The Databricks governance foundation for data, business context and AI assets. It provides fine-grained access controls, discovery, data lineage, automated quality monitoring, auditing and business semantics. With Unity Catalog, you get a single catalog across all enterprise assets, including structured data, unstructured data, models, agents, MCP servers, tools and skills.
- **Unity Gateway:** The AI control plane that extends Unity Catalog governance into every AI interaction at runtime. It governs AI usage, cost control, security, model choice and the interactions between models, agents and tools as they happen.

Together, they provide a unified governance layer for your data and AI, powering your enterprise agents. You govern data and AI together, rather than stitching together separate governance systems for data, models, agents and AI providers. With shared identity, permissions, policies and telemetry, you can create a consistent governance model that scales with your AI adoption.

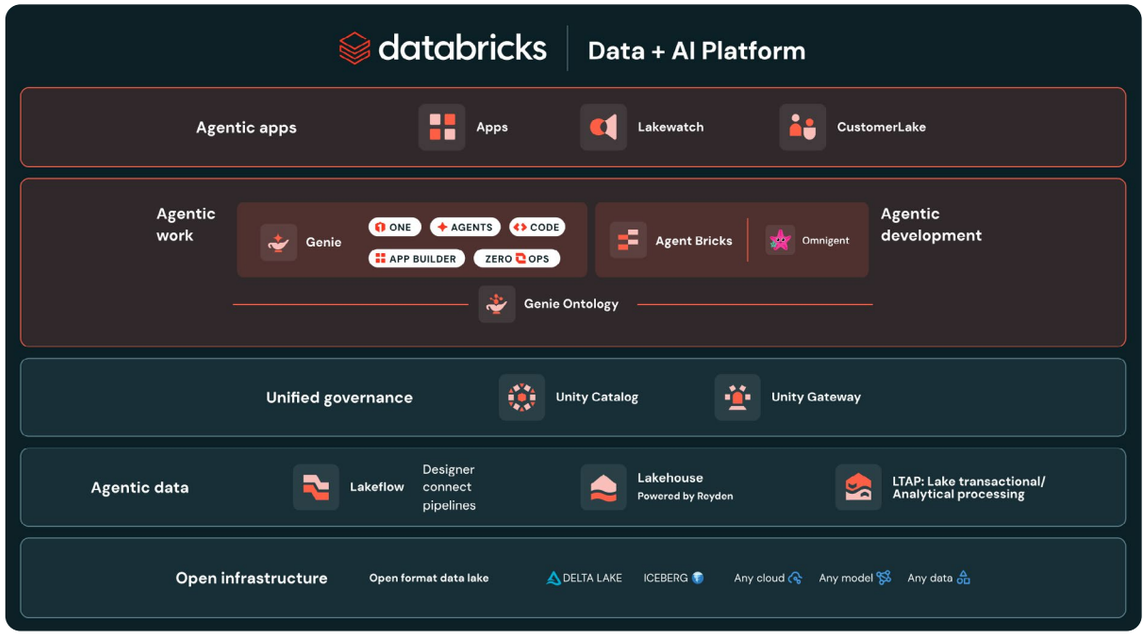


Figure 1. Databricks marketecture

Unity Catalog: Build the governance foundation for data and AI

Unity Catalog is the industry's only unified governance solution for data and AI across clouds, data formats and data platforms. More than 14,000 organizations govern their data and AI on Unity Catalog.

Unity Catalog is organized around three pillars:

1. **Secure and govern**
   Fine-grained access controls, including attribute-based access control (ABAC), let you define who can access what based on user attributes, data classifications and governed tags. Row filters and column masks protect sensitive data without duplicating it. Unity Catalog audits every action, and lineage tracks how data flows from source to consumption across both data and AI assets.
2. **Discover and understand**
   Unity Catalog provides a single place to discover, understand and trust your data and AI assets. Business semantics, including metrics, glossary definitions and domains, give AI systems the context they need to produce reliable results. Discovery and search make it easy for teams to find the right data, models and tools without having to navigate a maze of catalogs.
3. **Open and interoperable**
   Unity Catalog supports open formats and open protocols. OpenSharing enables secure, zero-copy data sharing across organizations and platforms used by thousands of enterprises. You avoid locking governance to a single engine or platform, preserving the flexibility to use the best tools for each workload.

Critically, Unity Catalog governs AI assets alongside traditional data assets. It registers models, agents, MCP servers, tools and skills as securables, inheriting the same permissions, lineage, auditing and discovery capabilities you already use for tables and files.

Unity Gateway: The control plane for AI governance and cost controls

While Unity Catalog governs what AI assets exist and who can access them, Unity Gateway governs what AI does when it runs.

Unity Gateway extends Unity Catalog governance across all AI models, providers, agents, coding agents, MCP servers and tools. It governs both Databricks-hosted AI and external AI through a consistent interface, so you don't need a different governance approach for every provider or framework.

Unity Gateway is organized around three primary themes:

- **Cost:** Understand, attribute, control and optimize AI spend. Track tokens, requests and costs across all models and providers. Set budgets and spend limits by user, team and application. Use Smart Routing to automatically select the most cost-efficient model for each task.
- **Choice:** Give teams access to leading proprietary and open models while maintaining flexibility as the frontier changes. Get day-zero access to new models, reliable capacity for production workloads and consistent interfaces that let you swap models without rebuilding applications.
- **Control:** Securely govern and trace AI interactions with centralized assets, identity-aware policies, guardrails and observability. Apply runtime policies based on identity, context and action. Capture end-to-end traces across models, agents, MCP servers and tools. Investigate and improve AI behavior over time.

Why customers choose Databricks to unify data and AI governance

The Databricks Platform is a modern unified platform for data, analytics and AI. It's built on open lakehouse architecture that brings together data engineering, data warehousing, data science, machine learning and AI application development in one place. Instead of stitching together point solutions, you consolidate your entire data and AI stack on a single platform with a single governance model.

That consolidation is what makes the Databricks approach to AI governance fundamentally different from the alternatives.

Governance is built directly into the platform

Unity Catalog isn't a standalone governance product that you layer on top of your infrastructure. It's the governance layer of the Databricks Platform — the same system that governs your tables, files and pipelines also governs your models, agents, MCP servers, tools and skills. When you build on the Databricks Platform, governance is already there. When you add AI workloads, they inherit the permissions, lineage, auditing, business semantics and access controls you've already defined for your data. There is no separate governance stack to deploy, integrate or maintain.

Unity Gateway extends that same governance into AI runtime

Because Unity Gateway is part of the Databricks Platform and built on Unity Catalog, every AI interaction (whether it involves a Databricks-hosted model, an external provider, a coding agent or an MCP server) inherits the same identity, permissions and policies that govern your data. You get cost controls, smart routing, model access, security policies, guardrails and observability through a single control plane that is already connected to your data, your business context and your enterprise identity.

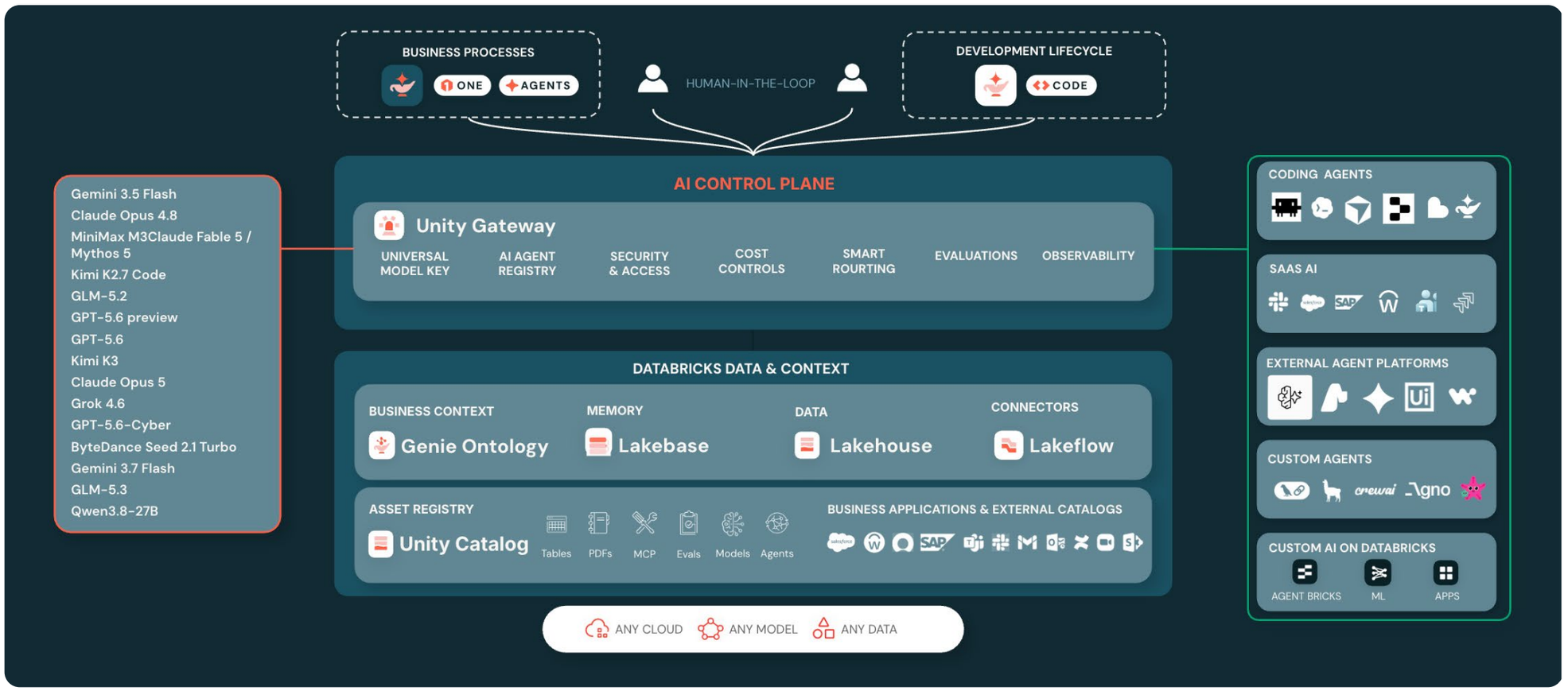


Figure 2: Databricks AI governance and management stack

One platform means one governance foundation

On the Databricks Platform, the path from raw data to governed dataset to business context to AI application to runtime AI interaction is continuous. Your data engineers, data analysts, ML engineers and AI engineers all work on the same platform, governed by the same system. That continuity eliminates the gaps that appear when you govern data in one tool, models in another, agents in a third and AI providers in a fourth. It also means your governance scales naturally as your AI adoption grows.

Open by design, no vendor lock-in

The Databricks Platform is built on open source and open formats (Delta Lake, Apache Spark™, MLflow and open table formats). Unity Catalog supports open protocols like OpenSharing for secure, zero-copy data sharing across organizations and platforms. Unity Gateway can also govern proprietary models, open source models, external providers and any agent framework not on the Databricks Platform through a consistent interface. You get the benefits of a unified platform without sacrificing the flexibility to use the best tools, models and providers for each workload.

Cost, choice and control are the outcomes of this unified approach. The chapters that follow show you how to achieve each one.

"Unity Gateway extends our Unity Catalog governance to external model endpoints, allowing us to remain coding-agent agnostic while giving engineers the freedom to use the tools that work best for them. At the same time, it provides centralized oversight and cost controls, with the ability to track AI spend from the individual engineer level up through entire departments."

— Bart Consedine, Head of AI Engineering, Magnite

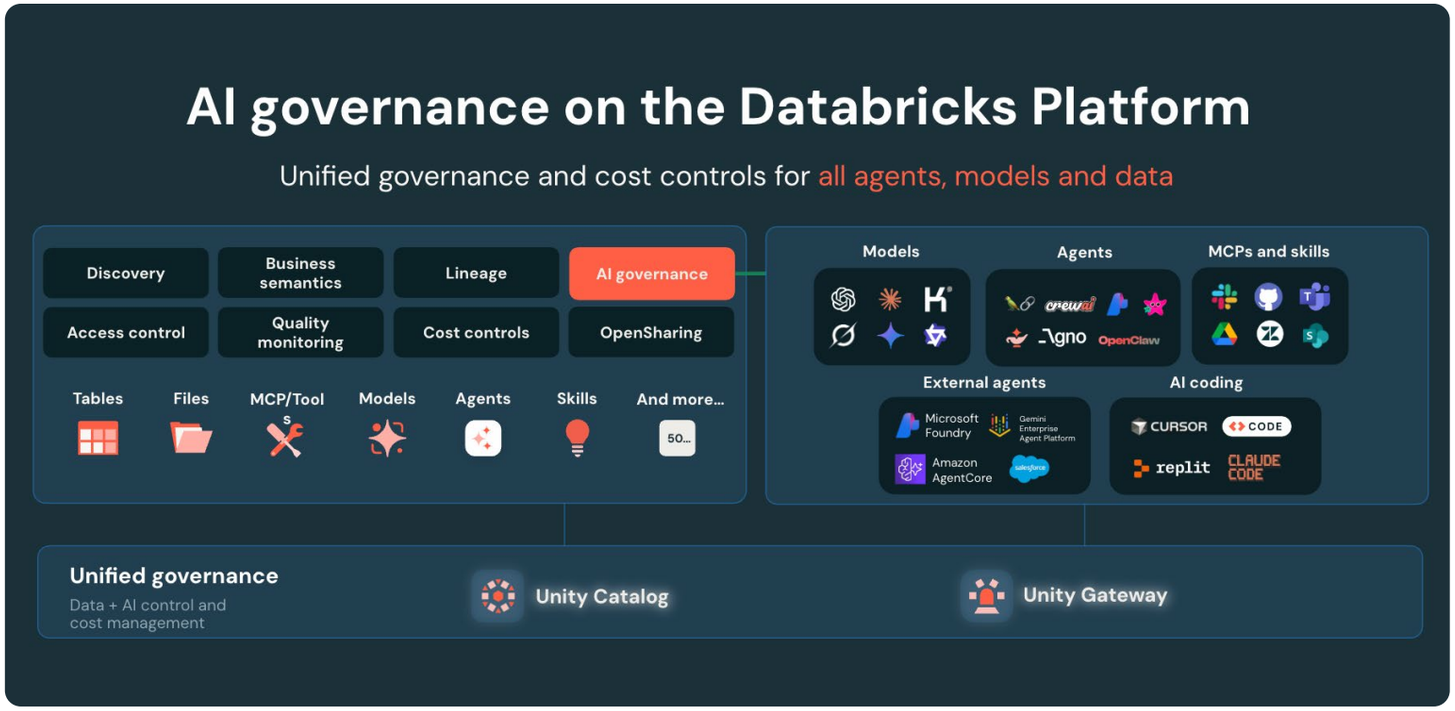


Conclusion

AI governance is no longer optional. It's foundational for making AI trustworthy, scalable and production-ready. By unifying data and AI governance on a single platform, you gain the visibility, control and flexibility to scale AI adoption with confidence. Unity Catalog provides the governance for your data and AI assets, while Unity Gateway extends that governance to every AI interaction at runtime.

The next three chapters show you how to put integrated, comprehensive AI governance into practice.

- **The "Cost" chapter tackles the economics of AI**: How to understand, control and optimize AI spend without slowing your teams down
- **The "Choice" chapter addresses model choice:** How to give your organization access to the best AI for every task while staying flexible as the frontier evolves
- **The "Control" chapter covers governance:** How to securely govern, trace and improve every AI interaction across your enterprise



02

Cost: Manage AI Costs Without Slowing Down Teams

Understand your AI spend

AI economics are fundamentally different from those of traditional software and cloud environments. With conventional infrastructure, you provision capacity and pay for compute time. With AI, you pay per token while token consumption is driven by the complexity of the task, the model you choose, the number of reasoning steps and the tools your agents invoke along the way.

Agentic workloads amplify this dynamic. A single agent session might involve dozens of model calls, tool invocations, retries and reasoning steps, all of which consume tokens. Coding agents are particularly significant drivers of variable usage because they operate continuously, generating and evaluating code across large code bases. When hundreds or thousands of developers use coding agents daily, token consumption can grow faster than anticipated.

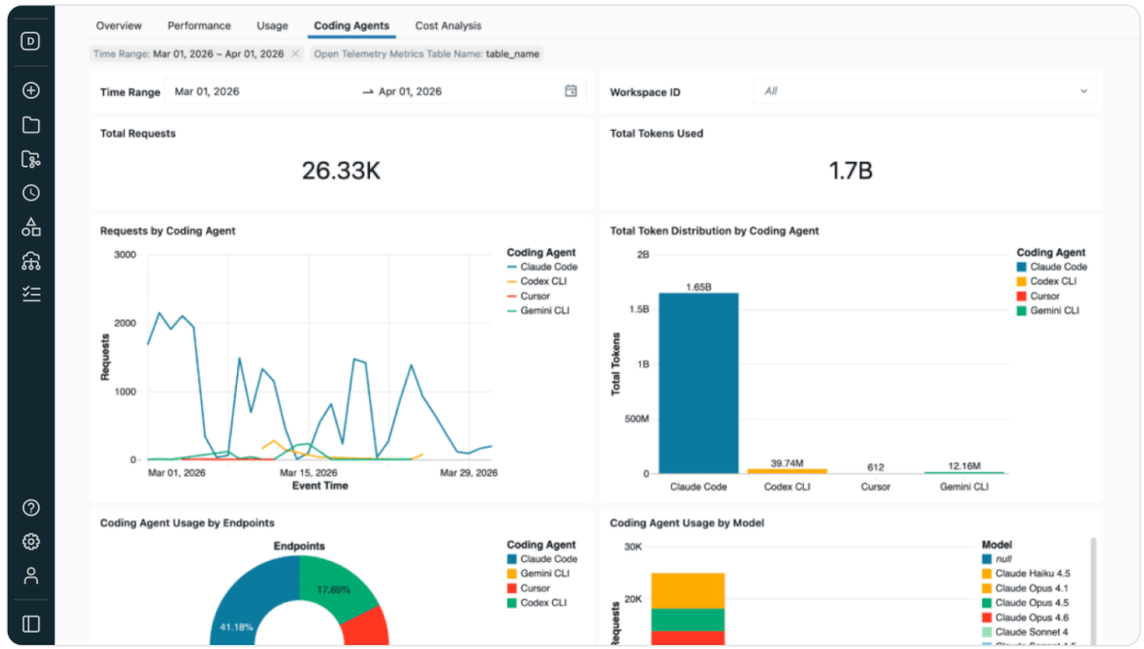


Figure 3. A Databricks developer dashboard showing active spend

Your first step is to **centralize spend visibility**. Instead of dealing with multiple isolated provider dashboards, you need to track tokens, requests and costs across all models and providers in one place. With Unity Gateway, you get centralized usage and cost dashboards that break down spend by model, provider, workspace and use case.

BEST PRACTICE

Attribute spend by user, team, application, agent and workload — including external model spend. When you can see that a specific coding agent consumed $50,000 in tokens last month, or that a particular team's agent is retrying failed tool calls and burning tokens in the process, you can take targeted action. Without attribution, cost management is guesswork.

Build guardrails around spend

Visibility is the foundation, but you also need mechanisms to prevent runaway costs before they happen.

Unity Gateway provides layered spend controls:

- **Budgets at the per-user and team/application level** — Provide the ability to set overrides for specific workloads that need higher limits
- **Alerts** — Notify administrators and developers when usage approaches defined thresholds
- **Hard spend limits** — Automatically block requests when budgets are exceeded, preventing surprise bills
- **Rate limits (queries per minute and tokens per minute)** — Prevent any single user or application from consuming disproportionate resources

It's equally important to give developers visibility into their own consumption. When engineers can see how much their coding agent costs per day and how that compares to their team's budget, they self-regulate. Accountability shifts from central IT policing usage to teams managing their own spend, which scales far better.

```
AI Gateway Budgets for [redacted]
  See go/aidevtools/quota/wiki for more info

  Monthly Budget
  ▓▓▓
```

Optimize, don't just restrict

Not every task needs the most expensive model. A simple code completion doesn't require the same model as a complex multistep reasoning task. The cost-per-token difference between frontier models and efficient alternatives can be 10x or more, but the quality difference for straightforward tasks is often negligible.

This is where Smart Routing changes the economics of AI.

Smart Routing automatically selects the best model and harness for each task, optimizing across cost, quality and performance. Instead of routing every request to the most expensive frontier model, Smart Routing matches the task to the most efficient option that meets quality requirements.

The results are significant. Smart Routing can match frontier quality at up to **30% lower cost per task** by intelligently selecting the right model for each request. Combined with spend visibility and developer accountability, organizations typically see **50% or more in total AI cost savings.**

Key Techniques in AI Cost Management

Lower-cost models	Smart routing	Spend controls	Context optimization
OSS + more efficient models	Route models + harnesses dynamically	Visibility, warnings + budget tripwires	Compaction, tool pruning + cache tuning
~50%+ SAVINGS	~30% SAVINGS	~10% SAVINGS	~10% SAVINGS

Routing is a smarter cost lever than limiting developer access. When you restrict which models developers can use, you reduce flexibility and potentially reduce output quality. When you route intelligently, you preserve quality and developer freedom while automatically reducing costs.

Think of it as “valuemaxxing”: maximizing the value you get per AI dollar rather than minimizing the tokens you consume.

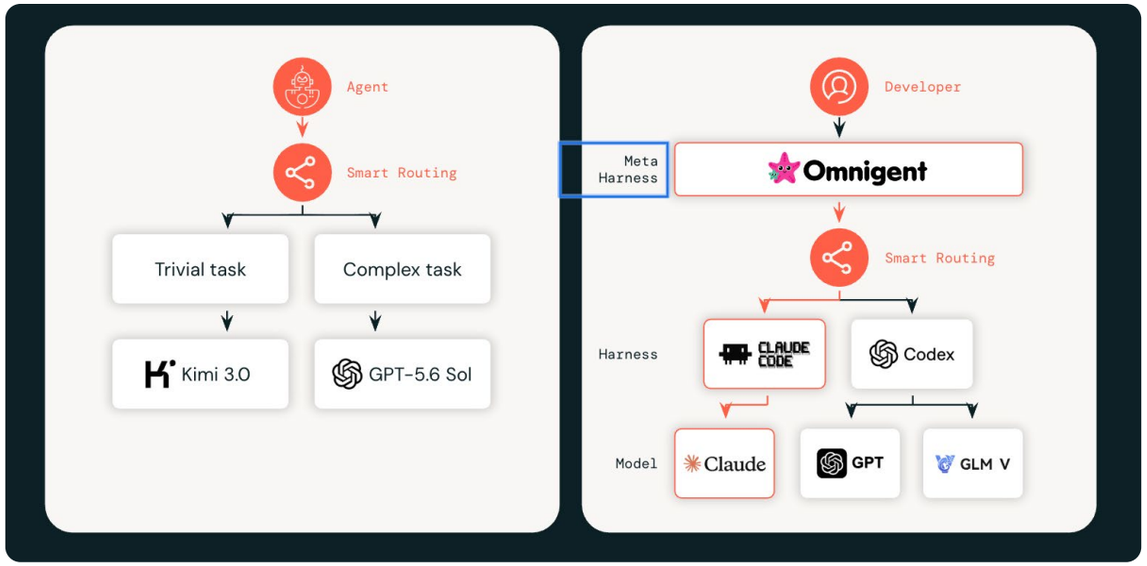


Figure 4. Smart Routing

BEST PRACTICE

Route work based on the task instead of standardizing the whole organization on one model. Let Smart Routing handle the optimization so developers focus on building rather than benchmarking models.

Further reading

- **Managing AI coding costs at scale** — A practical guide to understanding and controlling coding agent spend
- **Smart Routing in Unity Gateway: Match frontier quality at 30% lower cost per task** — How intelligent model selection optimizes cost without sacrificing quality

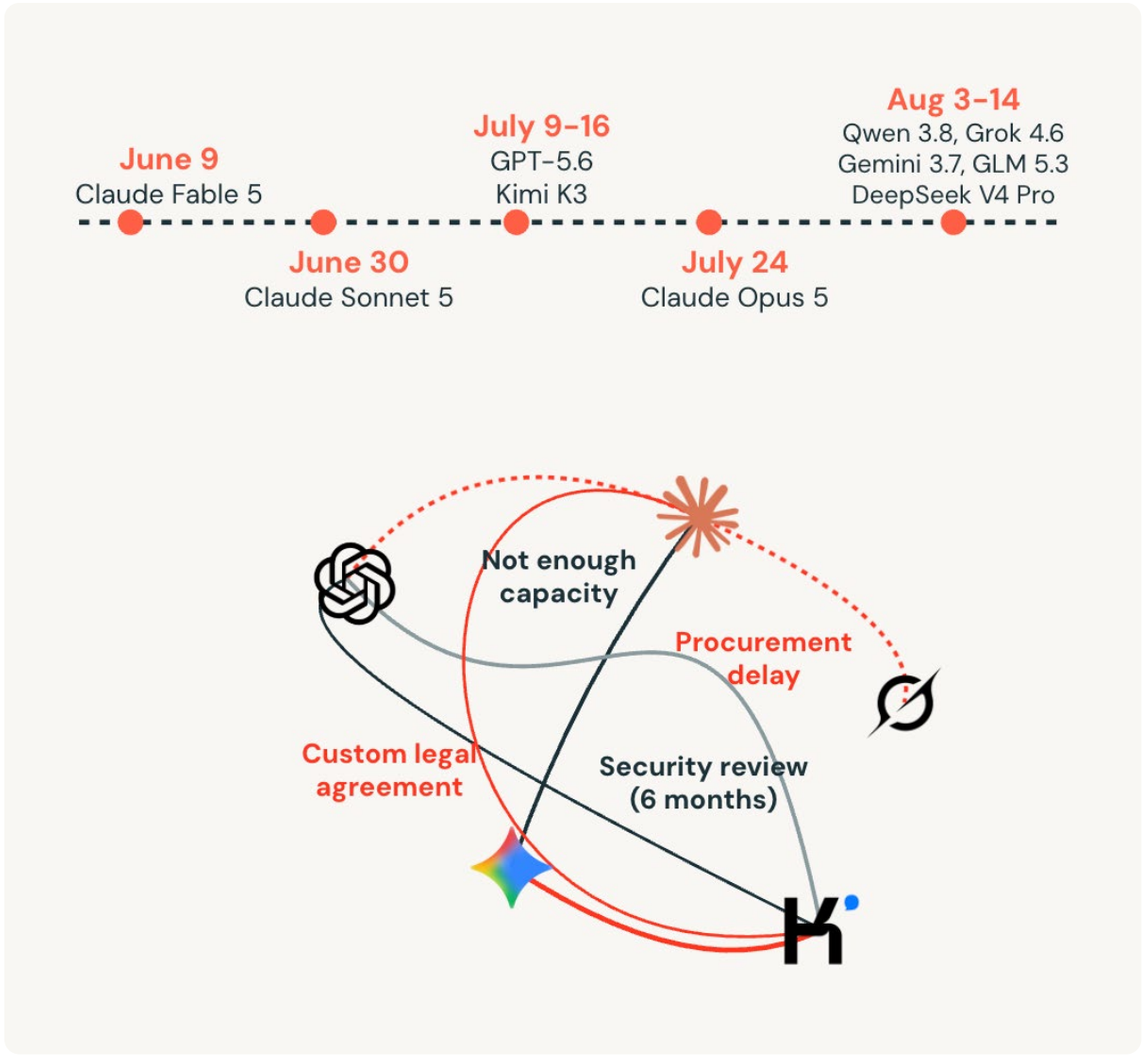
03

Choice: Give Your Organization the Best AI for the Job

Why model choice matters

There isn't one winning model. Different models excel at different tasks, whether it's code generation, reasoning, summarization, analysis, creative writing or domain-specific work. They all have different performance profiles across model families — the model that leads benchmarks today may be surpassed next quarter.

But that's not all. The proprietary and open source ecosystem is evolving at an extraordinary pace. New frontier models are launched, on average, every five days. Open models are increasingly competitive with proprietary alternatives across many tasks. Because some models are better suited to specific jobs, developers want access to multiple models and coding agents.



When you standardize on a single provider, you create architectural and commercial dependency. If your applications are tightly coupled to one model's API, switching costs are high. If your contracts lock you into a single provider's pricing, you lose leverage as the market evolves. And if that provider experiences capacity constraints or outages, your applications go down with it. You need flexibility.

Give teams broad, reliable model access

Unity Gateway provides access to leading proprietary and open models through foundation model APIs, a single, governed interface for your entire organization.

This means:

- **Day-zero access** to the latest frontier and open source models as they become available. Your teams don't wait weeks for procurement and integration.
- **Reliable capacity** backed by production-grade inference infrastructure that automatically scales with demand. More than 2,400 customers use three or more model families through Unity Gateway today.
- **Model provider services** that let you bring your own keys (BYOK) and expose existing provider relationships through the same governed gateway. That way, you have one governed way to expose AI across the organization — whether the model runs on Databricks or on an external provider.

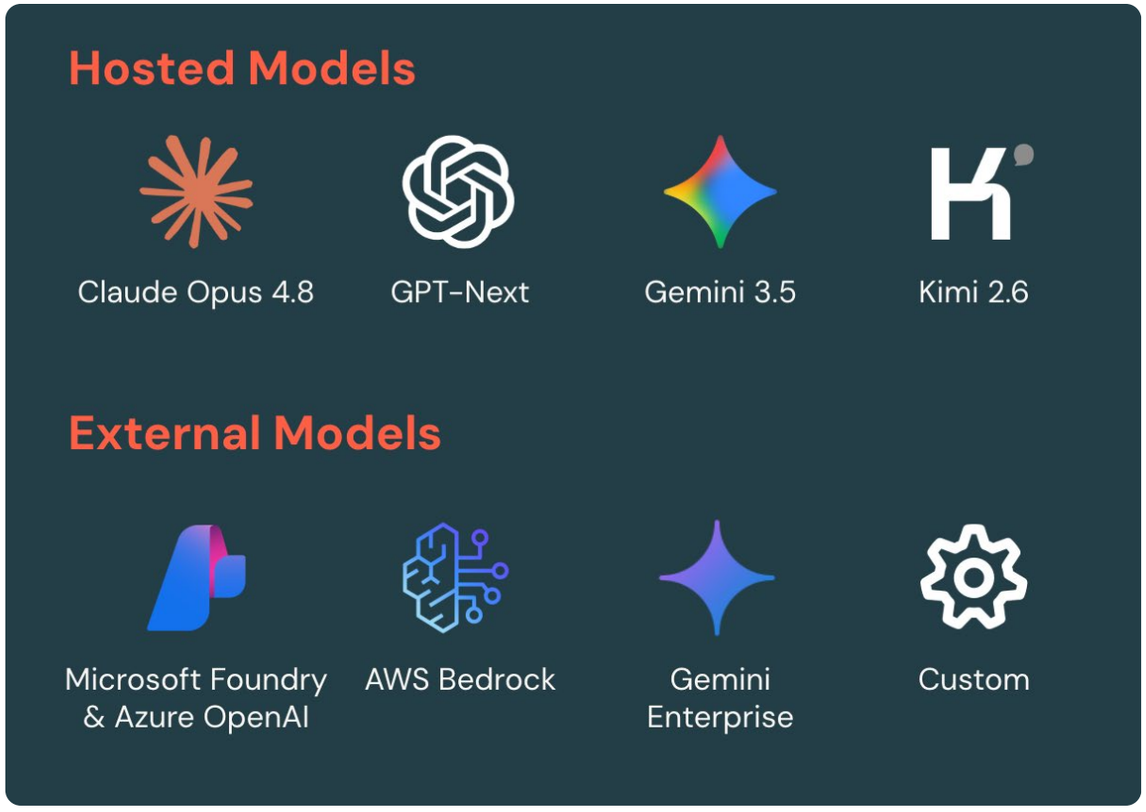


Figure 5. Unity Gateway lets you choose between OOTB or BYOM

BEST PRACTICE

Centralize access, not choice. Don't make every team stand up its own vendor relationship, credentials and governance stack just to get model choice. When you centralize access through Unity Gateway, teams get the models they need while the organization maintains consistent governance, cost controls and observability.

Stay flexible as the frontier and OSS landscape evolves

The model you choose today will not be the model you use a year from now. Your architecture needs to accommodate that reality.

Unity Gateway provides consistent, OpenAI-compatible interfaces across models and providers. This means you can swap the underlying model without rebuilding your application.

Additional resilience features protect your production workloads:

- **Traffic splitting:** Lets you gradually shift traffic between models, testing new options without risking production stability
- **Fallbacks:** Automatically route requests to alternative models when the primary destination is unavailable, throttled or experiencing latency
- **Session affinity:** Ensures that multi-turn conversations stay on the same model for consistency

BEST PRACTICE

Architect for substitution. Assume the preferred model will change, and make switching a configuration decision rather than an application rewrite. The organizations that move fastest on AI are the ones that decouple their applications from any single model.

Make choice intelligent

The “Cost” chapter discusses Smart Routing to optimize economics by selecting the most cost-efficient model for each task. Here, the lens shifts to making model selection dynamic and intelligent.

Your developers shouldn’t have to benchmark every new model that launches. They shouldn’t have to decide whether a task needs a frontier model or whether an efficient alternative would produce equivalent results. That decision should happen automatically.

Smart Routing auto-selects based on task complexity and capabilities, choosing the right model and harness combination and optimizing across quality, speed and cost. When a better model enters the ecosystem, it can be added to your stack without developers rewriting workflows. Smart Routing evaluates it against existing options and automatically routes appropriate tasks to it.

BEST PRACTICE

Automate model selection where possible so teams benefit from a changing frontier without taking on the complexity themselves. The goal is to make the best model for the job the default, without requiring every developer to become a model evaluation expert.

Further reading

- **Introducing Omnigent: A meta-harness to combine, control and share your agents** — How Omnigent enables flexible agent composition across models and frameworks
- **Benchmarking coding agents on the Databricks multimillion-line codebase** — Data shows that harness and model choice matter, and open models are competitive with proprietary alternatives

04

Control: Securely Trace Every AI Interaction

As AI moves from answering questions to taking action, you must govern not only which AI services people can access, but what models and agents can actually do at runtime. You also need visibility into every interaction so you can understand, investigate and improve AI behavior.

Govern your entire AI estate

The first step in controlling AI is knowing what AI exists in your organization: which models, agents, MCP servers, tools, connections and coding agents. All of these need to be registered, discoverable and governed.

Unity Catalog serves as the unified AI asset registry. Every AI asset is registered as a Unity Catalog securable, inheriting the same governance model used for tables, files and other data assets:

- **Permissions and EXECUTE grants** — Control who can use each AI asset
- **Attribute-based access control (ABAC)** — Lets you define access policies based on user attributes, data classifications and context
- **Identity-aware access** — Ensures that AI interactions respect the identity of the user on whose behalf the agent is acting — not just the service account running the agent
- **On-behalf-of-user identity** — Propagates the end user's identity through the entire chain of model calls, tool invocations and data access, so permissions are enforced consistently

That framework connects Unity Gateway back to the Unity Catalog foundation. The same identity, permissions and policies that govern your data also govern your AI.

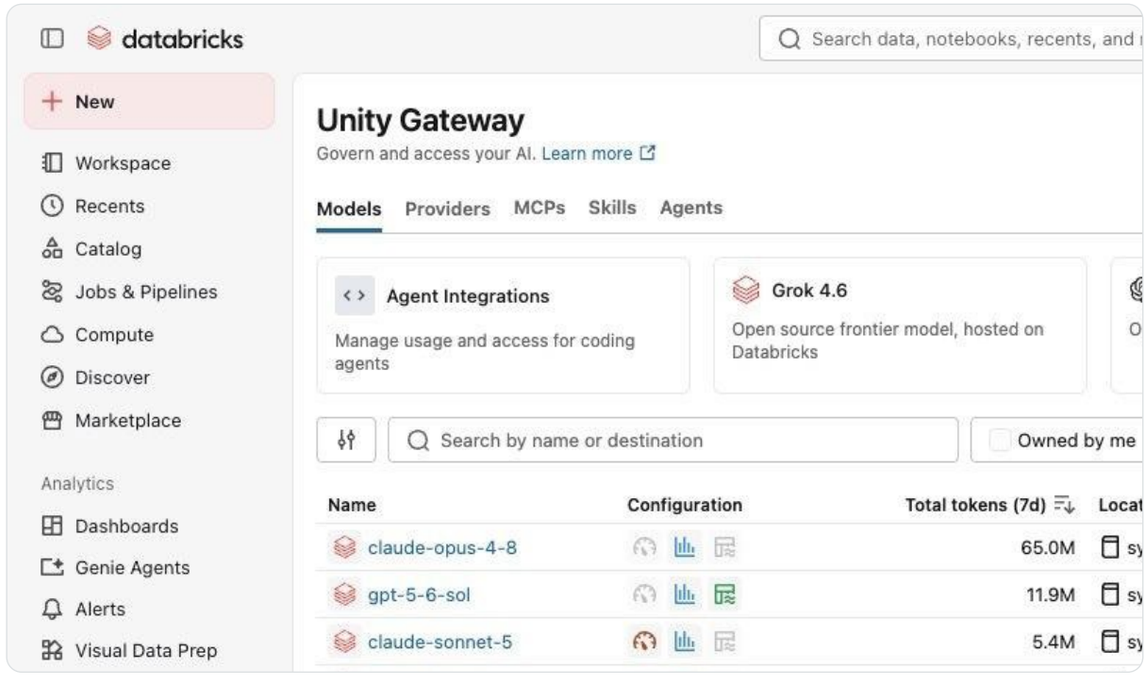


Figure 6. Unity Gateway allows you to control AI across your organization

BEST PRACTICE

Maintain a unified AI asset registry so that security and platform teams know which AI exists, who can use them and how they are connected. Shadow AI, which comprises agents and tools that operate outside your governance perimeter, is the fastest-growing risk in enterprise AI.

Govern AI behavior at runtime

Knowing what AI assets exist and who can access them is necessary but not sufficient. You also need to govern what AI does when it runs.

Unity Gateway introduces runtime governance capabilities that go beyond traditional access control:

- **Service policies:** Apply contextual controls to AI interactions — allowing, denying or requiring approval for specific actions based on identity, context and the action being requested. For example, you can allow an agent to read from a Jira MCP server but require approval before it pushes code to a GitHub repository.
- **Guardrails:** Protect against common AI risks in real time and include PII detection and masking, prompt injection defense, jailbreak prevention and unsafe content filtering. You can use built-in guardrails or integrate external guardrail providers.
- **Tool-level permissions:** Control which tools an agent can invoke and under what conditions

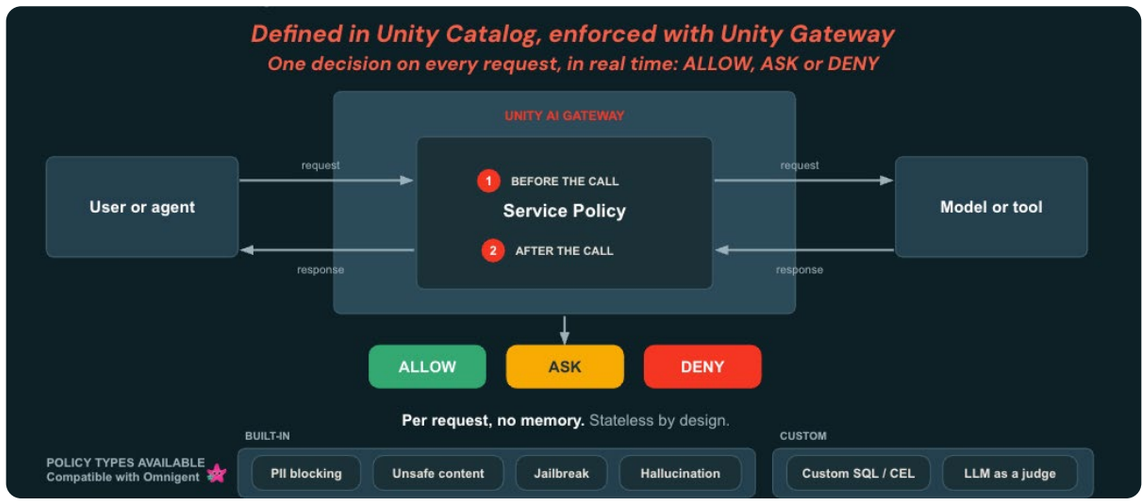


Figure 7. Govern AI actions, not just access

The key shift is from governing access to governing actions. Traditional governance asks "Can this user access this model?" Runtime governance asks "Should this agent be allowed to take this specific action, given this user's identity, this data classification and this context?"

BEST PRACTICE

Apply policies based on identity, context and action. A blanket "allow" or "deny" at the endpoint level is too coarse for production AI. You need policies that understand what the agent is trying to do and whether that action is appropriate given the circumstances.

Trace agent actions end to end

When something goes wrong with an AI system, like an agent accessing data it shouldn't, producing an incorrect result or taking an unexpected action, you need to understand exactly what happened. That requires end-to-end tracing.

Unity Gateway captures and securely stores AI interactions through multiple mechanisms:

- **MLflow Tracing:** Provides structured traces of model interactions, agent workflows and tool invocations
- **OpenTelemetry integration:** Emits standardized traces for every MCP tool call routed through Unity Gateway, including tool name, arguments, errors, token counts, latency and session IDs
- **Payload logging:** Captures the full content of prompts and responses for audit and investigation

Critically, trace data itself requires governance. AI traces often contain sensitive information: customer data in prompts, PII in responses, proprietary business logic in tool calls. Unity Gateway applies PII masking to trace data and stores traces in governed lakehouse tables, ensuring that your audit trail is both complete and protected.

Databricks engineers recently used Unity Gateway tracing combined with Genie to identify seven MCP tool bugs that were silently costing nearly $500,000 per year in wasted tokens and approximately 12,000 engineering hours per year in agent wait time. The entire process of finding, quantifying and fixing the bugs took about one hour. The issue was not writing the fix but knowing what to fix.

BEST PRACTICE

Treat AI traces as sensitive governance data, not disposable application logs. Your traces are the audit trail that proves what AI did, why it did it and whether it complied with your policies. Govern them accordingly.

Monitor, investigate and improve

Control is not just prevention. It's the ability to understand, investigate and improve AI behavior over time.

Unity Gateway provides centralized AI telemetry covering usage, performance, quality and risk. This telemetry flows into governed lakehouse tables, creating a single source of truth for AI operations.

From there, you can:

- **Track** AI usage patterns, cost trends, model performance and policy compliance across your entire AI estate

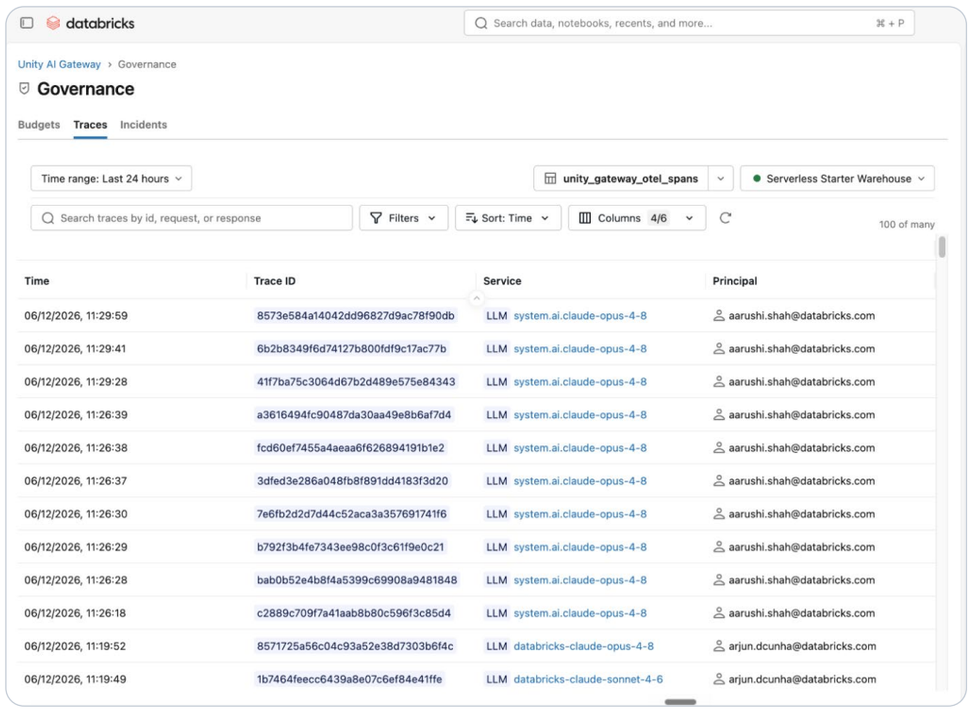
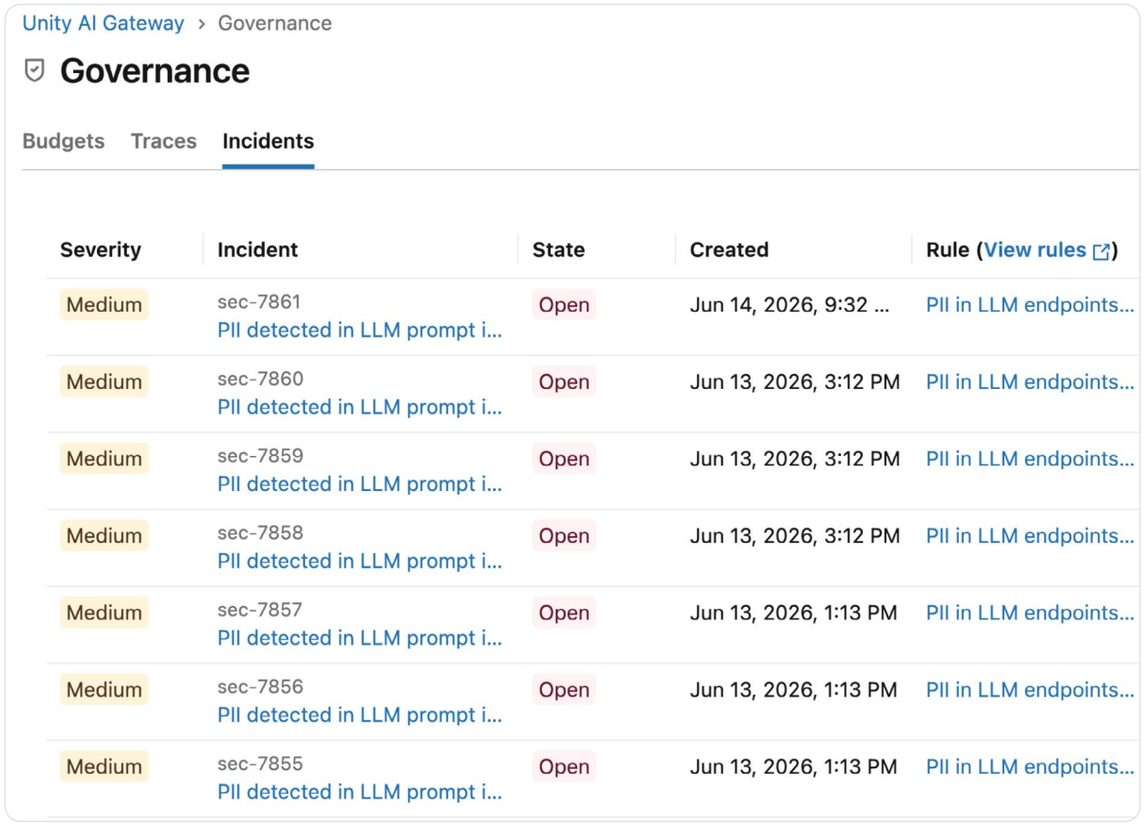


Figure 8. Single tracing table for all agent interactions across models and MCPs

- **Investigate** incidents using Lakewatch, the Databricks open, agentic SIEM, for SOC-level investigation and Genie for natural-language analysis of AI activity. When something goes wrong, you can ask questions like "Which tool errors recur the most?" or "What does each error cost in tokens and wait time?" and get answers in minutes rather than days.



- **Improve** AI operations continuously by using governed telemetry to identify optimization opportunities, such as inefficient model usage, failing tool calls and underperforming agents, and take targeted action.

BEST PRACTICE

Control is not just about prevention; it's also about understanding, investigating and improving AI behavior over time. The organizations that get the most value from AI governance are the ones that treat telemetry as a continuous improvement tool, not just a compliance checkbox.

Further reading

- **How to Safeguard AI Workloads With Unity Gateway Guardrails** — A practical guide to implementing guardrails for PII protection, prompt safety and content filtering
- **How we eliminated $1 million a year of wasted AI agent spend in one hour with Unity Gateway and Genie** — A real-world example of using tracing and Genie to find and fix costly agent tool failures

Getting started

AI governance is the key element and the foundation that makes AI trustworthy, scalable and production-ready. Organizations that govern AI effectively are the ones that scale AI fastest, because their teams can move with confidence rather than caution.

Databricks helps you unify data and AI governance on a single platform, giving you the **cost** visibility and optimization to make every AI dollar intentional, the **choice** and flexibility to adopt the best AI for every task without vendor lock-in and the **control** to securely govern, trace and improve every AI interaction.

The next step is yours.

05 Case Studies



ENSEMBLE™

CUSTOMER STORY

Ensemble scales governed AI with Unity Catalog

(source)

10,000 tables

Migrated to Unity Catalog in one night

$46 billion

In net patient revenue managed

8,000+ users

Enabled by upstream data engineering work

databricks

Ensemble provides end-to-end, technology-enabled revenue cycle management (RCM) for hundreds of hospitals and physician groups nationwide, leveraging unmatched healthcare data scale to help providers navigate rising payer pressure and thin margins. As Ensemble expands their AI initiatives across the revenue cycle, from patient engagement to account resolution, governance has become foundational.

The organization needed a unified approach to securely power AI while maintaining strict HIPAA compliance. To create a robust governance platform with better data discovery, cataloging, access control, lineage and auditing capabilities, Ensemble turned to Databricks and Unity Catalog.

Managing complex environments with limited capabilities

Ensemble supports approximately 8,000+ users, enabled by upstream data engineering work, including BI, operations and innovation teams that consume the data. The company also has a dedicated AI team designing and deploying multiple use cases across its data lake, including prior authorizations, coding and charge capture validation, accounts receivable follow-up, clinical denials and appeals, underpayments and build-up rate optimization.

To support these expanding AI initiatives, Ensemble is building a Unified Data Hub (UDH), a Platinum data layer designed to standardize enterprise data products across clients and business lines. The UDH serves as the governed foundation for AI and analytics, ensuring that trusted, well-documented datasets are consistently consumed across the organization.

Ensemble's legacy Hive Metastore environment proved difficult to replicate across user environments. Back-end processes were required to implement data masking, creating operational complexity. "We have a production workspace and a user acceptance testing (UAT) workspace, and it was really hard to move a copy of production data down to UAT. Having fresh, recent data is important," said Pheanouk Pel, AVP of Data Engineering at Ensemble.

Data discovery and cataloging capabilities were limited, and access control, lineage and auditing were not sufficient for Ensemble's needs. With healthcare data requiring strict governance and the growing demand for AI-driven use cases, Ensemble needed a more robust platform to consistently manage data across environments while meeting the security and compliance standards expected in healthcare.

Migrating to Unity Catalog with a custom framework

To address these challenges, Ensemble migrated from Hive Metastore to Unity Catalog. Preparation included generating a complete inventory of database objects and validating dependencies (particularly views) to ensure all referenced tables were migrated in the correct order.

"It was pretty painless," said Brad Beck, Senior Manager, Data Engineering at Ensemble. "We did a lot of validating — probably over-validated — in our test environment to make sure everything would work."

After several months of preparation, the full cutover to Unity Catalog was completed in a single night. The team migrated an estimated 10,000 tables. Ensemble updated their custom tooling platform, Flux, to support Unity Catalog. Because most pipelines are built on this framework, once the platform upgrade was complete, teams were able to migrate seamlessly. Backward compatibility allowed critical data assets, approximately 85% to 90% of enterprise datasets, to move first, followed by supplemental data.

While the migration was operationally significant, the larger impact was strategic: Unity Catalog became the centralized governance layer supporting Ensemble's data products, AI models and enterprise reporting.

Unity Catalog also simplified cross-environment replication. Instead of relying on complex workarounds, the team can now use deep clones across catalogs to move data between production and UAT, significantly reducing operational friction while maintaining governance controls.



Immediate improvements in governance and data management

With Unity Catalog in place, data discovery, cataloging, access control, lineage and auditing capabilities improved immediately.

"Being able to look back at the lineage of data to know where it was sourced through all the layers has been very helpful," said Beck. "We really ramped up cataloging right after the UC migration, so we have the majority of our enterprise-level data cataloged. That's been a great search tool for our analytics teams."

Access control also became more robust. "Access control is much stronger in Unity Catalog with how we cascade roles and manage access requests," said Michelle McDearmon, AVP, Data Engineering at Ensemble.

The team has implemented a scalable HIPAA-compliant masking pattern using Azure role-based access control (RBAC), Unity Catalog policies and tag-based masking (ABAC patterns). Fields tagged as sensitive or PHI are automatically masked for users outside authorized groups, ensuring compliance without manual intervention for each new schema or table.

Unity Catalog has also strengthened Ensemble's AI governance posture. The company is leveraging Agent Bricks to power use cases such as a predictive mapper that helps standardize incoming healthcare data from new clients. Because Agent Bricks operates within Unity Catalog's governed environment, AI agents inherit the same access controls and policies as the underlying data.

"When we use Agent Bricks, the agent becomes an object that we can permission just like anything else," said Pel. "That makes secure AI deployment seamless."

For Ensemble, this consistency ensures that AI innovation does not compromise the security of protected health information.

The team also leverages Unity Catalog system tables for cost monitoring and observability. Tags pushed down from external orchestration tools, such as Azure Data Factory, enable detailed dashboards that break down consumption by client, business line and team. This visibility allows Ensemble to track usage and make informed resource decisions.

Building the future of governed healthcare data products

Looking ahead, the Unified Data Hub (UDH) will play an increasingly central role in Ensemble's architecture. As enterprise data products are standardized into the Platinum layer, Unity Catalog will provide the governance backbone, ensuring consistent policies, lineage and access controls across analytics and AI workloads.

The next phase includes expanding metadata richness, strengthening documentation and integrating Microsoft Purview to enhance discoverability across business users. By formalizing enterprise data products within the UDH, Ensemble enables faster, self-service access to governed data across the organization, reducing bottlenecks and putting trusted data directly in the hands of those who need it.

With Unity Catalog as the control plane for data and AI governance, Ensemble is positioned to accelerate AI innovation while maintaining the security and compliance standards required in healthcare.



edmunds

CUSTOMER STORY

Edmunds: From lakehouse to digital mind

databricks

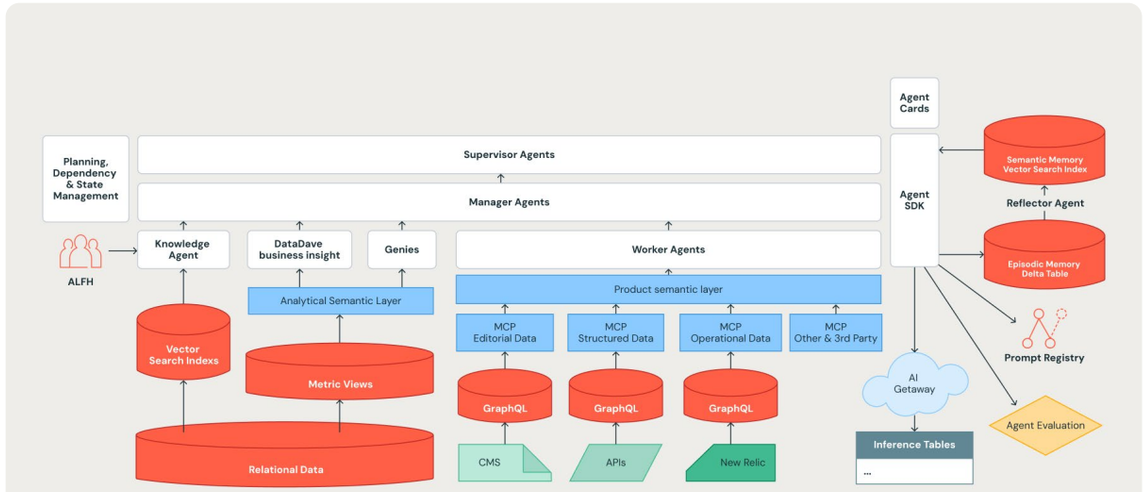
The challenge

Edmunds is one of the most trusted automotive information and shopping platforms in the United States. The company had built a rich data lakehouse on Databricks containing vehicle inventory, expert and consumer reviews, pricing intelligence, vehicle specifications, incentives, and media. This data is Edmunds' competitive advantage, its "data moat."

But the data was largely passive. Teams accessed it through dashboards and manual queries, and turning insights into action required human interpretation at every step. As AI adoption accelerated across the industry, Edmunds saw an opportunity to transform their lakehouse from a static data store into an active, intelligent system. The challenge was clear: How do you build a governed, multi-agent AI ecosystem where specialized agents can reason over enterprise data, take action, and continuously learn, without losing control of security, costs, or data quality?

The solution: Edmunds Mind on Databricks

Edmunds built "Edmunds Mind," a governed, multi-agent AI ecosystem on Databricks Lakehouse using Agent Bricks. Rather than deploying a single chatbot or monolithic AI application, Edmunds designed a hierarchical agent architecture that mirrors how an organization operates.



Supervisor agents: handle strategic planning, task decomposition, dependency management and escalation. Manager agents coordinate domain-specific workflows, delegating tasks to specialists and collecting results.

Specialist agents: perform focused tasks like VIN decoding, image analysis, pricing calculations, SQL generation and data correction. If a manager can't resolve a task, it escalates the full context, including failed attempts and prior observations, to a supervisor, which can re-plan or request human intervention.

Four agent systems illustrate the approach in practice:

1. **Automated data enrichment.** When a vehicle-data issue is detected, such as an incorrect color on a listing page, a supervisor agent triages the problem. A vehicle data manager then coordinates specialist agents for VIN decoding, image retrieval and color analysis. A data correction agent proposes the fix. Critically, a human data steward must approve the change before it reaches production. Unity Catalog validates permissions at every step, and the complete interaction is logged for audit and future learning.
2. **Knowledge Assistant**. A customer-facing RAG-based system that combines expert reviews, consumer reviews, vehicle specifications, transcribed video, pricing and incentives to answer automotive questions in the Edmunds brand voice. Rather than relying on a static model knowledge base, the Knowledge Assistant retrieves current, governed data with metadata filtering and recency prioritization to produce accurate, up-to-date answers.
3. **DataDave analytics agent.** A multi-agent analytics system built around a five-stage generate-and-critique workflow: triage, planning, code generation, execution and synthesis. A dedicated critique agent validates output at each stage, catching errors before they reach the user. Edmunds reports approximately 95% accuracy on complex analytics queries, with use cases like correlating website traffic and demographic data to identify underserved dealerships.
4. **Pricing specialist team.** Edmunds decomposed a monolithic pricing agent into a coordinated team of True Market Value, Depreciation and Deal Rating agents, orchestrated by a manager agent. The output is a contextual explanation of why a vehicle has that valuation, drawing on multiple pricing dimensions simultaneously.

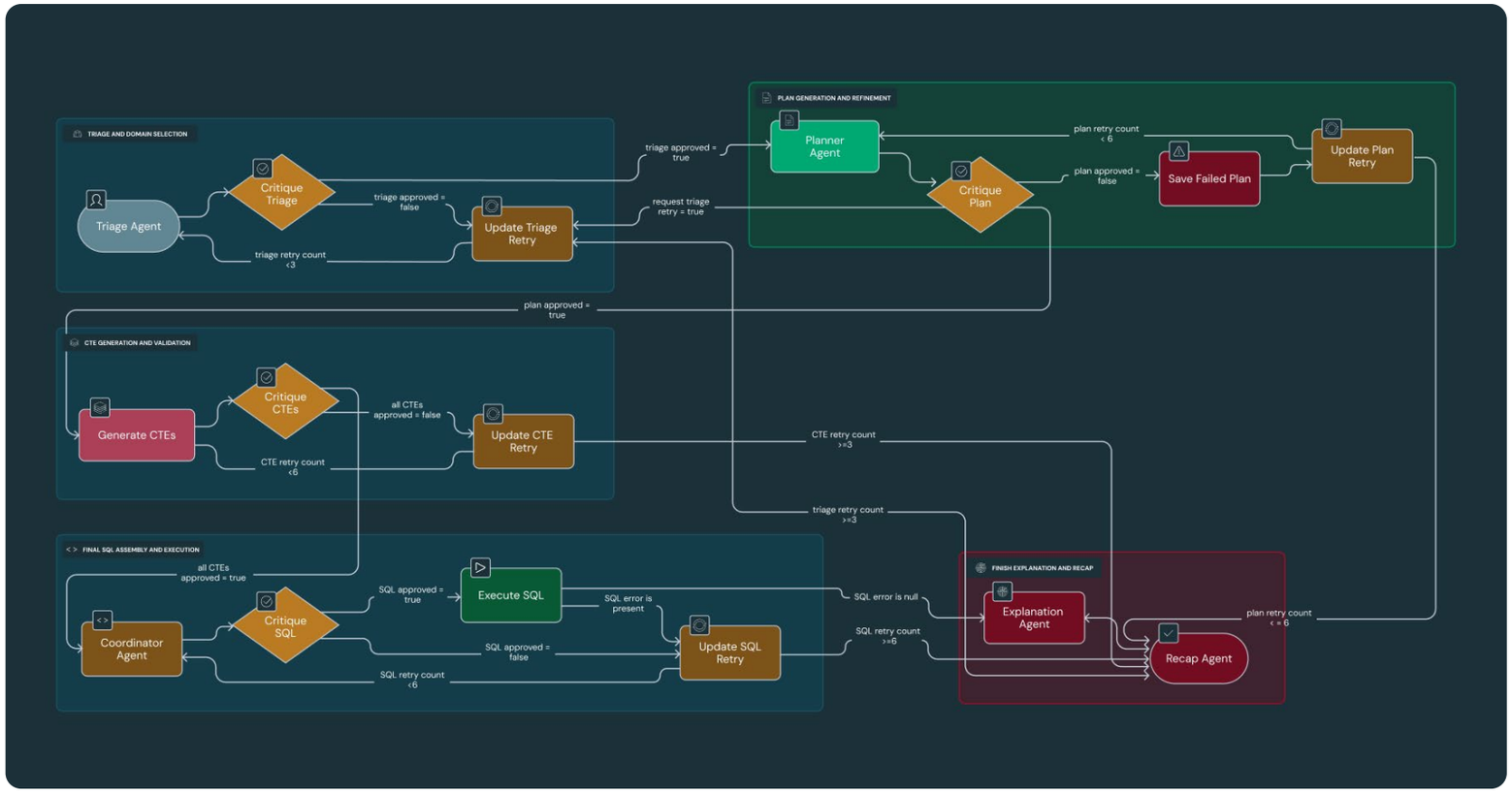


Figure 9. DataDave's "Generate-and-Critique" Workflow

Governance as the foundation

Governance is embedded throughout the Edmunds Mind architecture, not layered on after the fact. Three layers work together to keep agents productive and controlled.

Unity Catalog serves as the governance backbone, governing data, models, functions, tools, permissions, lineage and auditing across the entire ecosystem. Agents access enterprise data through governed semantic layers that are aligned with the same schemas used by the Edmunds website and dashboards. This alignment reduces semantic drift between customer-facing products, internal analytics and AI agents. Unity Catalog also ensures that agents can only access data and tools they are explicitly permitted to use.

AI Gateway (now Unity Gateway) provides runtime governance, including model routing for cost control, rate limits, observability and centralized tracing across model and tool calls. Edmunds uses this to match tasks to the right model: high-value reasoning tasks use stronger frontier models, while routine tasks use more cost-efficient alternatives. This keeps AI spend intentional and attributable.

Human-in-the-loop controls ensure that agents operating on high-impact workflows, like modifying production vehicle data, require human approval before changes take effect. Failures are explicitly escalated upward rather than silently retried. This design acknowledges that agents should augment human judgment on sensitive operations, not replace it.

Shared memory as an enterprise asset

One of the most distinctive elements of Edmunds Mind is its two-level memory system, which turns agent experience into reusable organizational knowledge.

Episodic memory records what happened during each agent interaction: actions taken, tool calls made, observations, failures, intermediate results and human approvals. Semantic memory stores generalized, reusable learnings extracted from those episodes, such as recurring anomalies, successful strategies or known failure patterns.

A background Reflector agent periodically reviews episodic events and consolidates useful findings into semantic memory. This creates a compounding feedback loop. Agents attempt work. Successes and failures are recorded. Reusable knowledge is extracted. Future agents retrieve that knowledge to perform better. Over time, the system gets smarter, not just from better models, but from accumulated organizational experience.

The result

Edmunds Mind represents a shift from reactive data consumption to proactive, governed AI operations. By building on Databricks Lakehouse with Unity Catalog for asset governance, Unity Gateway for runtime governance, and Agent Bricks for agent construction and orchestration, Edmunds created an AI ecosystem where specialized agents collaborate on complex tasks, learn from experience, and operate within consistent governance controls.

The organization is also democratizing AI development through Agent Bricks, enabling domain experts across the company to build and deploy agents without deep engineering expertise. An internal "LLM Agents Guild" supports engineers and citizen developers alike, accelerating adoption while maintaining architectural consistency.

> "Databricks lets us run the right model for the right task–securely and at scale. That flexibility powers our agents and delivers smarter car shopping experiences."
>
> — Greg Rokita, VP of Technology, Edmunds



novo nordisk

CUSTOMER STORY

Building more impactful clinical trials with generative AI

(source)

20% of the team uses clinical data

With the FounData platform, they can access billions of data points

$157M+ in net new value

Optimized clinical trials are expected to boost value over the next five years

$14M in operational efficiency gains

With Databricks, Novo Nordisk has greatly accelerated drug development

databricks

Novo Nordisk, a global healthcare and pharmaceutical leader, is on a mission to defeat serious chronic diseases. But internally, Novo Nordisk faced challenges in efficiently managing and accessing it's vast clinical trial data due to siloed systems — obstacles that hindered collaboration, slowed research timelines and created bottlenecks in decision-making. With the adoption of the Databricks Data + AI Platform, Novo Nordisk transformed its approach to data, enabling seamless collaboration, robust data governance and AI-driven insights. In just nine months, they moved from concept to MVP, unlocking the ability to scale machine learning models, automate previously manual workflows and design better trials based on real-world evidence — all while ensuring compliance with rigorous regulatory standards. Crucially, over 20% of Novo Nordisk's development organization now actively uses clinical data. It's expected that over the next five years, this will result in more than $157M in net new value and over $14M in efficiency gains attributed to optimized clinical trials. With a shared data foundation now in place, teams across the enterprise can act faster, learn from one another and ultimately bring life-saving treatments to patients sooner.

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Siloed patient data impedes healthcare advancements

Novo Nordisk, a century-old global healthcare company headquartered in Denmark, is dedicated to transforming patient lives by addressing diabetes, obesity and chronic diseases, delivering swift access to life-changing treatments and the best possible care.

As Christian Sørensen, Head of Automation and Digital Innovation at Novo Nordisk, puts it, "Patients are waiting — how do we go faster?" This question guides every decision at Novo Nordisk, and it reflects the company's commitment to leveraging data and innovative technologies like generative AI to accelerate the pace of discovery and patient impact. To fulfill their mission, Novo Nordisk has developed multiple platforms to harness the power of data and AI across their clinical and research activities.

Their FounData platform is designed for the secondary use of clinical trial data once the trials are complete. FounData democratizes access to the billions of data points generated from Novo Nordisk's clinical trials, allowing scientists, medical experts and analysts to derive insights and make informed decisions, such as identifying and targeting suitable patient populations for clinical trials based on demographics, medical history and disease prevalence. Christian described FounData as an environment where "collaboration and back-and-forth exchanges between scientists and medical experts lead to deeper insights." With FounData, researchers can efficiently explore data, leading to faster and more informed research discoveries.

While FounData focuses on post-trial data, the DataCore platform is used to manage data throughout the clinical trial conduct. This platform operates in a GCP (Good Clinical Practice) environment, ensuring adherence to stringent regulations for data quality, security and compliance throughout the trial process. GCP compliance is crucial for ensuring the integrity and reliability of clinical trial data, which can later be seamlessly transferred to FounData for secondary analysis and interpretation.

According to Christian, both platforms are part of a broader, enterprise-wide transformation aimed at scaling the use of AI and reducing the time it takes to bring new treatments to patients. "This is our top strategic play for how we run clinical development. It's all hands on deck, including upskilling and growing our AI acumen." Sid Prabhu, Senior Director and Head of FounData, added that the value starts with data quality. "There's recognition from senior leadership that good quality data enables better, faster decisions," and that the quality of AI outcomes "improves dramatically when built on top of high-quality, scalable data."

With clinical trial data stored across various systems, it was challenging for teams to locate and utilize the data efficiently. Christian explained that "not a single person has knowledge of all datasets," and previous systems lacked the governance and accessibility necessary to streamline collaboration. Their existing platforms were fragmented, and siloed data created bottlenecks, preventing the full potential of their AI initiatives from being realized. Christian described the legacy environment as "a patchwork spiderweb. We had enterprise solutions, but they weren't widely adopted. So individual storage solutions were mushrooming across the org, leading to inefficiencies." This fragmentation created significant knowledge gaps. Only certain people knew what they were doing, insights were trapped in silos and information had to be passed manually between teams. This led to the loss of context and slower outcomes.

Another challenge was maintaining data integrity. "We had prerequisites for choosing a platform. We needed a platform that could leverage multiple coding languages. It also needed to be robust enough to handle complex data transformations while ensuring data lineage, tracking every step from collection to analysis. Data integrity, quality and testing were important," explained Christian. Novo Nordisk also required a solution that could handle structured and unstructured data types, which was critical for the diverse use cases they wanted to explore, particularly in clinical trials and research.

As Christian emphasized, the previous model placed the burden on individuals to compile data from multiple sources. "We expected that every data consumer had a head of integration on the team. Now we're serving data directly to users from one point of integration." This shift to a centralized foundation was critical to enabling AI at scale across the organization.

Driving compliance and research collaboration with Databricks

By choosing the Databricks Data + AI Platform, Novo Nordisk integrated various data sources, ensured seamless governance and provided a collaborative platform that fosters innovation across their data and AI projects. Databricks helps by enabling robust data pipelines, maintaining lineage and ensuring that insights generated in FounData can be traced back to their original sources in DataCore. This ensures both compliance and accuracy.

Novo Nordisk leverages various components of the Databricks Data + AI Platform to enable their data-driven initiatives, ensuring scalability, security and flexibility across their AI and research processes. One of the most important aspects of Databricks is Unity Catalog, which plays a central role in governance, ensuring that data access rights are strictly enforced and that every dataset can be traced to its origin. As Christian explained, "Data lineage is incredibly important for us. With FounData and DataCore, we need to know exactly where the data originated and maintain compliance with stringent regulatory requirements. The ability to trace everything back to the source is essential."

Novo Nordisk is also using Databricks Agent Bricks to deploy AI Agent Systems. While large language models (LLMs) are powerful, they are most effective when customized to perform specific tasks using relevant data. AI Agent Systems involves the use of AI agents that specialize in particular areas, working together like a well-coordinated team to solve complex problems. Novo Nordisk refers to this system as a Co-Scientist, where these AI agents not only analyze clinical data but also consult external resources, like medical literature, to provide comprehensive insights. This approach amplifies the effectiveness of Novo Nordisk's research efforts while ensuring that human oversight and collaboration remain central to the process.

With Agent Bricks Model Serving, the organization can securely deploy machine learning models, including large language models at scale, ensuring they meet compliance and security requirements. According to Christian, "The advantage we see with Model Serving is compliance and security — particularly for confidential data. We know who is using the model and why, down to a granular level." This feature allows Novo Nordisk to confidently manage sensitive clinical data and ensure that its models are properly governed.

Databricks also facilitates collaboration across Novo Nordisk's diverse teams. With the Workspaces feature, scientists, medical experts and data teams can collaborate within a unified environment. "The back-and-forth between scientists and medical experts in a workspace is what leads to truly valuable insights," said Christian. "By using AI, they can find, access, connect and solve." Workspaces help streamline the iterative research process.

Christian emphasized the strategic importance of sticking to Databricks' reference architecture. "We're not doing a highly customized setup. What we've built is essentially a clinical data repository on an industry-agnostic platform," he explained. "That's rare — and it's because we've had a strong partner in Databricks. It's helping us scale efficiently beyond clinical development."

Christian noted that Databricks now powers foundational components of collaboration and reproducibility. "We've adopted the medallion architecture and shared responsibility model: who builds what, who owns what, how we team around data. These are core to how we now operate, and we've removed duplicate effort," Christian emphasized. From a user enablement perspective, Databricks also supports standardization and templatization. According to Sid, "We have a repeatable blueprint now. You onboard your data to DataCore, set access via Unity Catalog, spin up infrastructure and build your front end. That's made it much faster to stand up new initiatives."

As Christian explained, Databricks plays a critical role in balancing speed with risk in a regulated environment. "It is OK to innovate. It is OK to experiment. But when we take our products into production, we ensure we're moving forward well — because the cost of mistakes is simply too high."

In addition to Model Serving and Unity Catalog, Novo Nordisk also extensively uses Databricks Lakehouse and Apache Spark™ Declarative Pipelines. "We rely heavily on pipelines, jobs, and the data sharing tools that underpin everything we've built," said Christian. "The success here has been using the Databricks toolbox effectively, pushing it to evolve in ways that solve real business problems."

Databricks AI/BI Genie is also being used as part of Novo Nordisk's early agentic AI implementations. AI/BI Genie enables business users, including those without technical or clinical backgrounds, to query and explore patient data in natural language. "Imagine you join Novo Nordisk tomorrow," said Christian. "It's fairly easy to make data-driven decisions because we can interact with the data without having the in-depth 30-year industry experience." The Co-Scientist system, supported by Agent Bricks and AI/BI Genie, searches across Novo Nordisk's clinical data foundation and returns curated, context-aware insights within seconds. Users can quickly surface information about adverse events, patient subpopulations and trial protocols — functionality that previously required weeks of manual work. They launched their first Co-Scientist agent in mid-June 2025.

Built on top of Databricks, AI/BI Genie respects defined access controls and data lineage while supporting collaborative exploration. “With the use of our strong foundation on Databricks, I can ask AI/BI Genie: ‘Show me all of our obesity data,’ and it pulls the relevant trials, insights and visualizations instantly,” Christian explained. AI/BI Genie plays a critical role in unlocking both data access and user empowerment, making AI-powered exploration a default capability across Novo Nordisk’s clinical development teams.

Underpinning all of this is a hybrid Data Mesh architecture built on Databricks’ lakehouse architecture. While domains like research, development and product supply each own their strategic data via decentralized DataCore nodes, governance is unified across the enterprise using Unity Catalog and data contracts. “We will never meet the demand of our company if we don’t design for self-service,” said Jonatan Selsing, Principal Platform Architect. “So we established a federated model where domains ingest their own data, and we empower them to build what they need.”

This approach allows for consistent regulatory compliance while accelerating research. Novo Nordisk now operates across eight cloud regions, with over 60 Databricks workspaces and more than 60 teams building in production. Standardization via Terraform, Azure DevOps and GitHub enables secure and reproducible workflows, even in GCP-regulated environments. As Jonatan put it, “Technology is no longer the bottleneck. Now our focus is on improving people, process and quality.”

Democratizing data for improved clinical trials

Databricks has contributed to the democratization of data across the organization, breaking down silos and enabling cross-functional teams to collaborate more effectively. By providing medical experts, data scientists and commercial analysts access to a single platform, the ability to derive insights has vastly improved. This has led to enhanced collaboration and the faster discovery of insights that were previously locked away in different parts of the organization.

Novo Nordisk has experienced significant improvements in how it manages and utilizes clinical trial data. One of the most remarkable achievements is the rapid time to value. “In just nine months, we were able to move from concept to MVP,” explained Christian. This accelerated timeline is supported by Databricks’ flexible, integrated platform, allowing Novo Nordisk to build on their existing infrastructure with minimal disruption. “This is one of the fastest digital projects we’ve ever completed," Christian adds, highlighting the platform’s ability to bring tools like Azure and Databricks into their environment seamlessly.

Novo Nordisk now has 1,000 users actively engaging with FounData, accessing, analyzing and collaborating on data in ways that were previously impossible. There are currently over 80 live collaborations, with AI/ML solutions being developed across multiple domains. “We’re delivering value well above our initial investment,” said Sid, “both in terms of productivity and in the future impact of what we’re unlocking.”

That level of impact stands out in a landscape where most data and AI initiatives struggle to deliver results. "It's really, really hard to point very concretely at business benefits," admitted Christian. "That's why we're focused on linking outcomes to real patient and process improvements." In contrast, Novo Nordisk has surpassed the industry benchmark of sub-10% ROI on AI projects by tying platform investments directly to speed, safety and scalability.

Christian emphasized that "over 20% of our development organization now actively uses clinical data via FounData — compared to just a fraction before, something like 1%." Christian added that DataCore has been scaled across all areas of development, supporting nearly all critical data sources. "We're now backing every major digital initiative in the development portfolio. The platform is officially endorsed as the enterprise-wide standard."

The introduction of Databricks has enhanced productivity and efficiency and resulted in faster data analysis and decision-making. By combining data from multiple clinical trials, the organization can now gain more specific insights into patient needs across various medical indications. "We're now sharper across more indications, which means we can build our clinical trials more efficiently and reach more patients," Christian mentioned.

Christian underscored the significance of this shift, particularly in a heavily regulated industry: "If we can automate documentation across the value chain, we can significantly cut back on lead times. That means faster delivery of innovation to patients." He emphasized that the impact of AI must be measured not just in algorithms, but in patient outcomes and process changes. "It is the change in end-user behavior that drives optimization in the investments we're making."

Novo Nordisk is also closing the loop between clinical trials and early-stage research. "We can simulate how molecules respond in the body and understand disease development earlier," Christian explained. "That data can inform how we design the next treatment." This data from SELECT now informs the majority of the early development pipeline, turning downstream evidence into upstream advantage and helping make drug discovery more circular.

The broader impact isn't limited to R&D. As Christian explained, "We're seeing cross-collaboration, like teams in Chemistry, Manufacturing, and Controls (CMC) using data to forecast clinical supplies. That data used to live in silos. Now it's shared by default." Looking ahead, Novo Nordisk is continuing to expand its use of AI agents, something that Christian views as a natural progression. "We're starting by democratizing the insights, not just the data. That's what makes the Co-Scientist so powerful," said Christian. "It's not just analyzing data, it's helping colleagues explore what's possible."

With the foundation in place and the Databricks Data + AI Platform as a core enabler, Novo Nordisk is well-positioned to accelerate the discovery of life-saving treatments, improve patient outcomes and continue pushing the boundaries of data-driven research.



Accolade | CUSTOMER STORY

Creating seamless healthcare experiences from start to finish

(source)

databricks

> Accolade time to insight has greatly improved, thanks to our ability to integrate complex data sources using Databricks
>
> - Kapil Ashar, Vice President, Enterprise Data and Clinical Platform, Accolade

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Accolade is a healthcare technology company that helps their members lead healthier lives by providing them with timely access to the right care. Challenged with fragmented data, Accolade partnered with Databricks to consolidate their data into a unified lakehouse architecture, which allowed the business to leverage large language models (LLMs) using retrieval augmented generation (RAG) techniques. With the help of generative AI, the Accolade team developed solutions to quickly answer complex member inquiries, bolstering major productivity gains.

Delaying AI efforts due to data fragmentation

Accolade, a healthcare company that provides care delivery and advocacy services, is on a mission to help every member lead their healthiest life. The company's technology-enabled solutions combine virtual primary care and mental health support, expert medical opinions and care navigation. Accolade's data group sought to connect the fragmented world of healthcare data in the name of better health. Spread across multiple platforms, Accolade's data was siloed and lacked real-time access, hindering accurate member stratification (i.e., the process of categorizing members into different groups based on health status, risk levels or healthcare needs) and timely care delivery.

Managing multiple data systems led to inefficiencies and increased operational costs for Accolade. According to Accolade's Vice President of Enterprise Data and Clinical Platform, Kapil Ashar, "Fragmented data across multiple different platforms hampered our efficiency and ability to provide timely, personalized care. Our internal teams struggled to quickly access the necessary information, leading to slower response times and a diminished ability to assist our members effectively." Accolade recognized the vital need to fix their data, remedy their challenges and enhance their service delivery, which could now be done with the help of generative AI. However, the complexity of managing and coordinating multiple models across various systems, such as AWS Redshift and Snowflake, proved to be a significant obstacle.

Since the company's complex data management landscape also complicated the development of AI-driven initiatives, numerous teams wanted to collaborate more efficiently and innovate at the pace needed in the competitive healthtech market. Therefore, Accolade's first step was to transition from a traditional BI infrastructure to an AI-driven framework.

Unifying data to create better customer experiences

Accolade leveraged several Databricks Data + AI Platform components to solve their data fragmentation, ML model development and customer service challenges. Central to this implementation was a lakehouse architecture that enabled the combination of data storage and management to facilitate easier access and analysis. This unified environment allowed Accolade to break down the silos between different types of data and systems, making it possible to harness data for their desired artificial intelligence use cases. Ashar elaborated upon the transition: "Databricks served as the foundational step for us. It's not just about having data; it's about making it actionable and efficient, which Databricks enabled us to do effectively at a scale we previously couldn't manage."

At the core of these real-time analytics, the Databricks Data + AI Platform tapped into Apache Spark™ for streaming data capabilities, enabling continuous data ingestion from various sources. As streaming data flowed into the system, Databricks Unity Catalog was critical for management and governance, supporting HIPAA compliance requirements with stringent access controls and detailed data lineage. Now, Accolade could seamlessly govern their structured and unstructured data, machine learning models, notebooks, dashboards and files across any cloud or platform. This unified approach to governance accelerated data and AI initiatives, simplified regulatory compliance and allowed data teams to access data and collaborate securely.

Databricks tools also paved the way for Accolade's use of retrieval augmented generation (RAG), a generative AI workflow that uses custom data and documents to provide context for LLMs. Accolade used the Agent Bricks Custom Agents to develop a RAG solution specifically tailored to improve the efficiency and effectiveness of their internal teams. The Databricks Data + AI Platform allowed the company's power users to access diverse data sources — from PDF files to online protocols and customer information — providing them with the guidance to answer questions confidently and correctly.

Accolade's RAG solution relied on DBRX, an open source LLM developed by Databricks, to help them use proprietary enterprise data for more accurate and context-aware outputs. For example, Accolade leveraged DBRX to enhance internal search functions and enable internal users to retrieve accurate information quickly. With the help of GenAI to analyze customer contracts and other relevant data, the internal teams could better understand and fulfill customer commitments. Once the DBRX model generated insights, Databricks Model Serving came into play. It deployed the model as a RESTful API, enabling real-time predictions that could be integrated directly into Accolade's decision systems. With Model Serving, the healthcare technology company had Databricks' support for version control, automatic scaling and integration for comprehensive lifecycle management. As a result of the Databricks integration, Accolade experienced greater transparency and control over resources and costs to simplify the development and maintenance of ML models, while ensuring compliance with data governance standards.

Centralizing data and AI tools for continued innovation

Databricks has enabled Accolade to integrate data and AI and catalyze innovation in customer service excellence. Consolidating team members' work in one place fosters a collaborative environment conducive to the rapid development and deployment of new technologies. Looking forward, AI and machine learning will continue to revolutionize how Accolade operates across various dimensions of their business. By democratizing access to high-quality insights, Accolade's compound AI system is not just enhancing productivity but also empowering employees to perform their roles with greater efficacy. The integration of RAG tools continues to support Accolade's internal teams, providing them with the confidence to handle complex queries — even if they lack expertise in a certain area. Accolade anticipates that the combination of human and artificial intelligence will remain core to their mission to deliver superior healthcare services to all.

Build trusted enterprise AI — see how

As AI agents, coding assistants and enterprise AI applications become part of everyday work, governance must extend beyond data to the runtime decisions AI systems make. Learn how to provide trusted business context, govern AI interactions, control AI spend and maintain flexibility across models, agents and clouds with a unified approach to data and AI governance.

Watch a Unity Gateway demo

Explore Unity Gateway

Try Databricks free

About Databricks

Databricks is the data and AI company. More than 20,000 organizations worldwide — including adidas, AT&T, Bayer, Block, Mastercard, Rivian, Unilever and 70% of the Fortune 500 — rely on Databricks to build and scale data and AI apps, analytics and agents. Headquartered in San Francisco, with 30+ offices around the globe, Databricks offers a unified platform that includes Lakebase, Genie, Agent Bricks, Lakeflow, Lakehouse and Unity Catalog. To learn more, follow Databricks on LinkedIn, X, YouTube and Instagram.

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